

Enhancing Solar Power Plant Efficiency with SVG



1. Introduction

Solar power plants have seen significant growth in recent years as an environmentally friendly and sustainable energy source. However, with the increasing penetration of diversified clean energy resources, optimizing active power and reactive power of these plants is crucial to ensure the stability of energy supply and harness their full potential. One key aspect of achieving this optimization is the implementation of Static Var Generators (SVGs) to stabilize the grid voltage. In this article, we will explain the concept of SVG and how SolaX C&I on-grid inverters can be utilized with integrated SVG functionality, leading to improved power quality and enhanced grid stability.

2. Understanding SVG

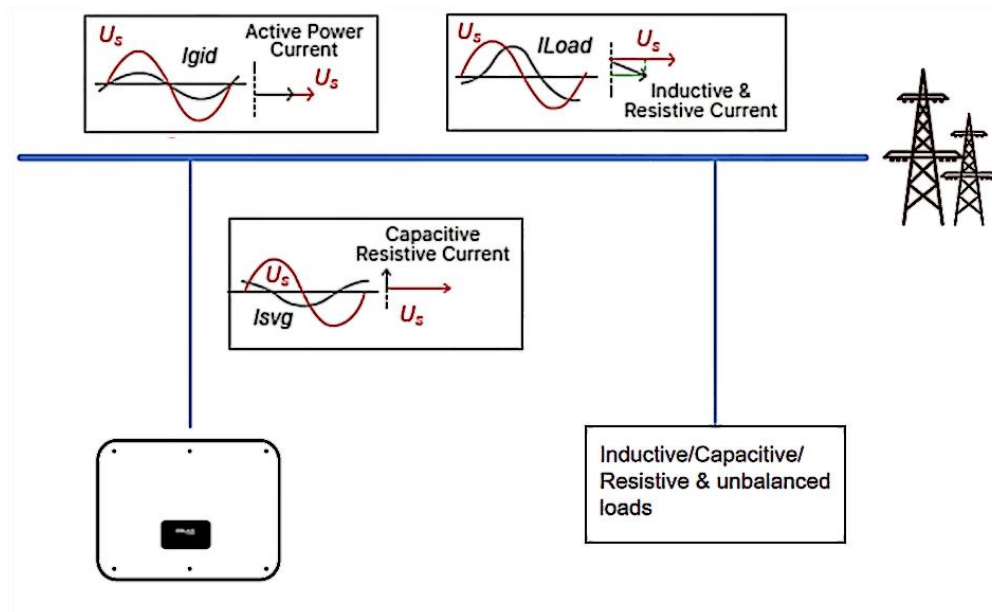
Static Var Generator (SVG) is a power electronics-based device that provides dynamic reactive power compensation in various applications. In solar power plant applications, SVGs are used to regulate and control the flow of reactive power in the electrical system. Reactive power is an important aspect of power systems that is necessary for voltage control and maintaining system stability. It consists of three basic functional parts: sensors, controller and compensation output module. By analyzing the current information measured by the outer CT or PT sensors, its controller will send Var compensation power to help regulate

voltage levels and power factor in the electrical energy system, ensuring efficient power transmission and stable operation. By rapidly absorbing or injecting reactive power as required, SVGs mitigate voltage fluctuations, minimize line losses, and improve overall power quality.

Overall, SVGs play a crucial role in reactive power compensation in solar power plant applications, ensuring optimal performance and grid stability.

3. SolaX Inverters with Advanced Software-Controlled SVG

By integrating SVG capability into solar inverters, SolaX inverter system operators can achieve dynamic or fixed reactive power compensation without the need of additional hardware like conventional SVG devices. Comparing with these SVG devices, SolaX inverters can play the rival role in power factor management for better power quality. This solution not only helps to save the initial investment costs of extra hardware devices, but also enables simply&easy installations and precise control of power factor and voltage levels, allowing solar inverters to dynamically respond to grid requirements.



The followings are benefits of SolaX inverters with integrated SVG functionality:

A. Improved Power Quality

With software-controlled SVG, solar inverters can actively regulate reactive power and power factor, reducing voltage fluctuations and harmonics. This significantly enhances power quality, ensuring smooth and stable operation of the electrical system.

B. Efficient Energy Delivery

By compensating for reactive power locally at the solar inverter level, line losses are minimized, resulting in more efficient energy delivery. This optimizes power transmission and reduces energy waste.

C. Grid Stability and Fast Response

During grid disturbances or fluctuations, SVG functionality allows solar inverters to provide voltage support and help stabilize the grid. This dynamic response aids in maintaining grid stability and mitigating power disruptions.

D. Scalability and Cost-Effectiveness

Integrating SVG functionality into solar inverters eliminates the need for separate SVG equipment. Hence, it simplifies system design, reduces installation costs, and improves scalability for solar power plants of various sizes. Even one of unit failed, the total reactive compensation power of the whole system will not be affected.

Conclusion:

SolaX inverters with SVG functionality can maximize the performance and efficiency of solar power plants, offer an effective solution for reactive power compensation. By actively regulating power factor and voltage levels, SolaX inverters enhance power quality, improve energy delivery efficiency, and support grid stability.

4. How to set the parameters

4.1 Advanced SVG integrated inverters:

Inverter Series	Firmware
X3 MEGA G2	All versions
X3 FORTH	All versions

Considering a better economic effect of the photovoltaic power plant system, the SVG function usually will only be activated at night to start compensation mode. Please set Reactive Power mode first. Reactive power mode includes "Fixed Q Power" for a fixed value of reactive power compensation and "PF Mode" for a fixed Power Factor.

4.2 Set Reactive Power on APP

Activate the function on 'SOLAX CLOUD' App:

Settings → ***Advanced Settings*** → ***Reactive Power Control***

4.3 Set Reactive Power on LCD

Activate the function by LCD screen:

Settings → **Advanced** → **GridServices** → **Reactive Power**

4.4 Enable SVG

Activate SVG function in the settings:

Settings → **Advanced** → **Check** → **SVG**

Note: Installer password is required after selecting "**Advanced**".

Once enabled, the SVG function will be activated automatically at night when there is not enough PV power. It's essential to note that the selection of a control mode depends on various factors such as grid characteristics, operational requirements, and regulations.